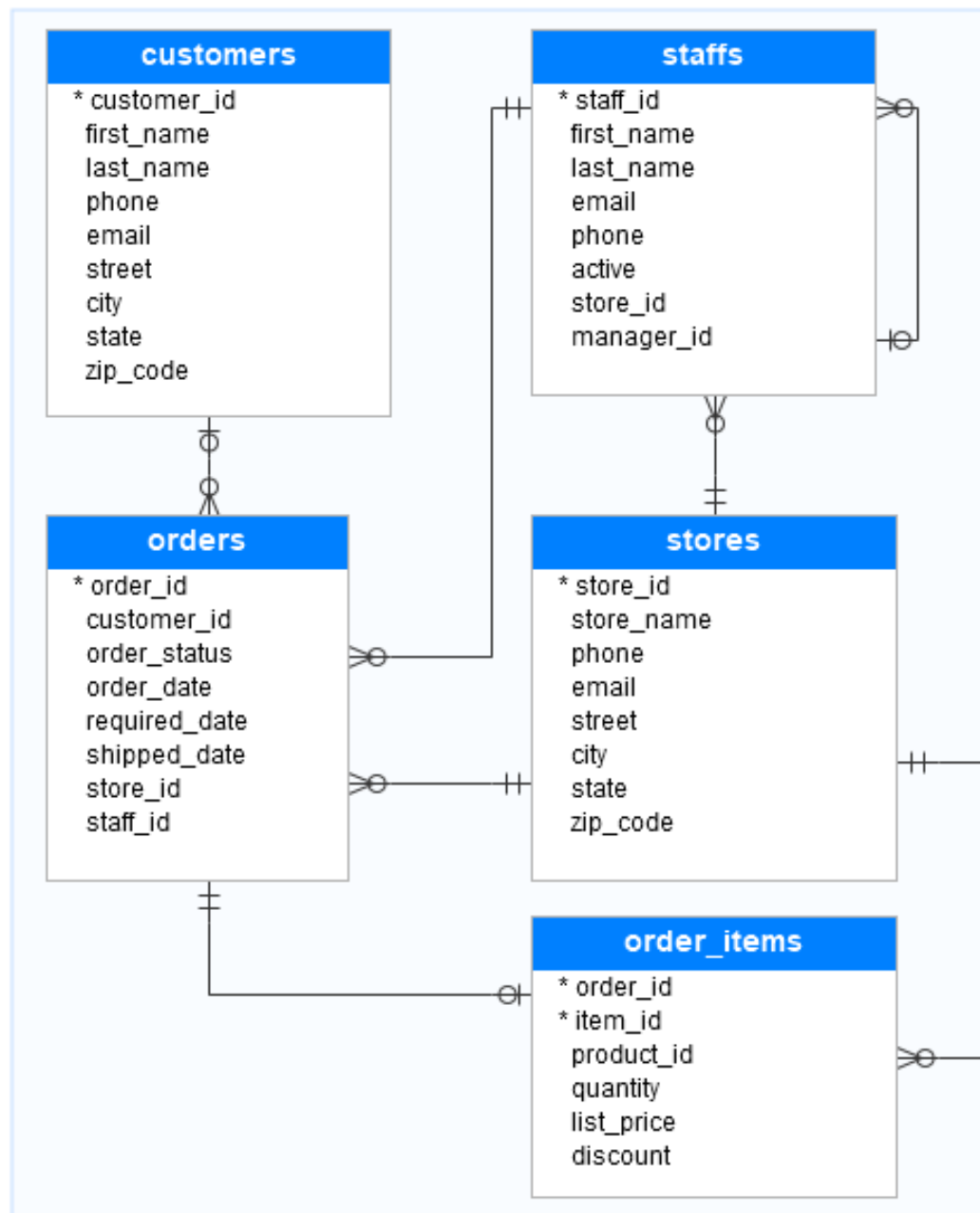




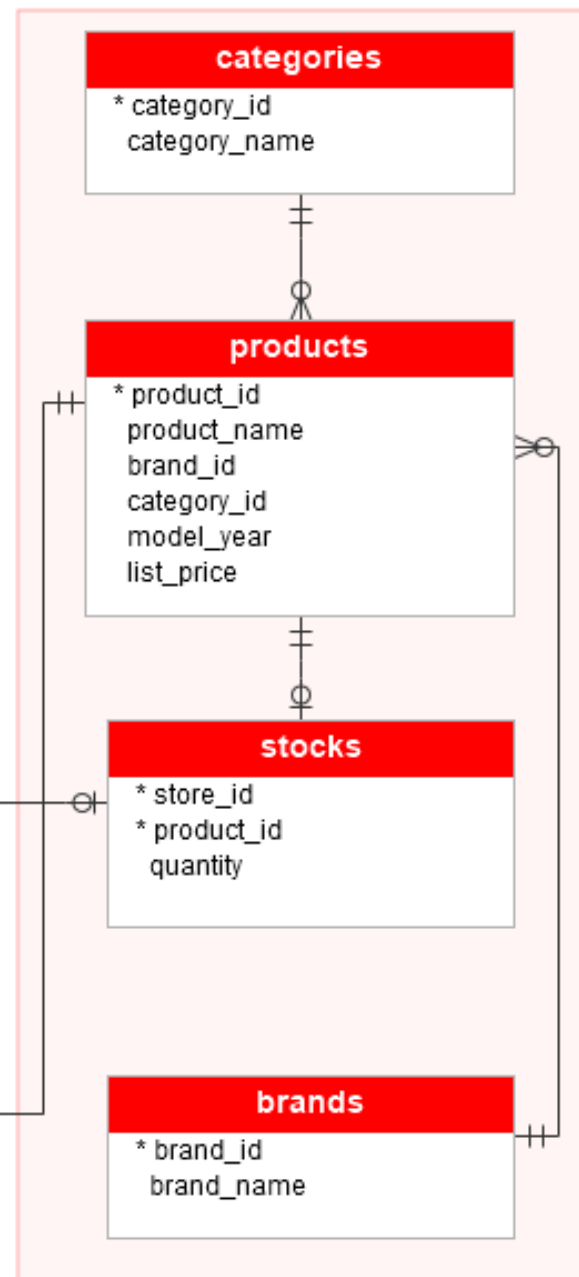
Raspodijeljene i nerelacijske baze podataka

VJEŽBA 02

Sales



Production



Zadatak

- Napraviti bazu podataka
 - **Postaviti:** „DB owner: sa”
- Preuzeti BikeStores.zip
- Otpakirati u jednu mapu
- Proučiti sadržaj datoteka
 - Koji jezik je u pitanju (SQL->DCL, DDL, DML)?
 - Koje datoteke i kojim redoslijedom trebamo izvršiti?
- Nakon izrade BP napraviti dijagram

Zašto „sa”?

SQL - ponavljanje

```
SELECT *  
FROM [NazivTablice]  
WHERE [FiltriranjePodataka]  
GROUP BY [KriterijGrupiranja]  
HAVING [FiltriranjePoAgregatnojFunkciji]  
ORDER BY [KriterijZaSortiranje]
```

1. Jesu li sve navedene ključne riječi obavezne za izvršavanje upita?
2. Je li redoslijed pojavljivanja ključnih riječi ispravan?
3. Koji se sve operatori mogu koristiti za filtriranje podataka?

SQL - ponavljanje

```
CREATE TABLE NazivTablice  
(  
  NazivStupca tipPodatka,  
  NazivStupca2 tipPodatka,  
);
```

Mogu li stupci imati isti naziv u jednoj tablici?
Mogu li se ponavljati nazivi stupaca u različitim tablicama?

```
INSERT INTO NazivTablice (imeStupca1, imeStupca2 , ...)  
VALUES (Podatak1, Podatak2, ...);
```

```
UPDATE ImeTablice SET  
stupac1=vrijednost1,  
stupac2=vrijednost2,  
...  
WHERE NazivStupca=NovaVrijednost;
```

Moraju li se podaci filtrirati (*WHERE*)
kod UPDATE i DELETE izraza?

```
DELETE FROM NazivTablice  
WHERE NazivStupca=Vrijednost;
```

SQL – spajanje tablica

- Ako želimo dohvatiti podatke iz više tablica trebamo koristiti JOIN
 - INNER JOIN
 - RIGHT JOIN
 - LEFT JOIN
 - FULL JOIN

	product_name	brand_name	category_name
1	Ritchey Timberwolf Frameset - 2016	Ritchey	Electric Bikes
2	Heller Shagamaw Frame - 2016	Heller	Cruisers Bicycles
3	Electra Townie Original 21D - 2016	Electra	Children Bicycles
4	Electra Cruiser 1 (24-Inch) - 2016	Electra	Children Bicycles
5	Electra Girl's Hawaii 1 (16-inch) - 2015/2016	Electra	Children Bicycles
6	Electra Moto 1 - 2016	Electra	Children Bicycles
7	Electra Townie Original 7D EQ - 2016	Electra	Children Bicycles
8	Pure Cycles Vine 8-Speed - 2016	Pure Cycles	Cyclocross Bicycles
9	Pure Cycles Western 3-Speed - Women's - 2015/2016	Pure Cycles	Cyclocross Bicycles
10	Pure Cycles William 3-Speed - 2016	Pure Cycles	Cyclocross Bicycles
11	Electra Townie Original 7D EQ - Women's - 2016	Electra	Children Bicycles

Zadatak:

Napraviti upit koji vraća podatke o nazivu proizvoda, branda i kategorije kojoj proizvod pripada.

SQL – spajanje tablica

```
select p.product_name,  
b.brand_name,  
c.category_name  
from production.products p, production.brands b, production.categories c;
```

Je li ovakav upit ispravan?

	product_name	brand_name	category_name
1	Ritchey Timberwolf Frameset - 2016	Ritchey	Electric Bikes
2	Heller Shagamaw Frame - 2016	Heller	Cruisers Bicycles
3	Electra Townie Original 21D - 2016	Electra	Children Bicycles
4	Electra Cruiser 1 (24-Inch) - 2016	Electra	Children Bicycles
5	Electra Girl's Hawaii 1 (16-inch) - 2015/2016	Electra	Children Bicycles
6	Electra Moto 1 - 2016	Electra	Children Bicycles
7	Electra Townie Original 7D EQ - 2016	Electra	Children Bicycles
8	Pure Cycles Vine 8-Speed - 2016	Pure Cycles	Cyclocross Bicycles
9	Pure Cycles Western 3-Speed - Women's - 2015/2016	Pure Cycles	Cyclocross Bicycles
10	Pure Cycles William 3-Speed - 2016	Pure Cycles	Cyclocross Bicycles
11	Electra Townie Original 7D EQ - Women's - 2016	Electra	Children Bicycles

```
select p.product_name,  
b.brand_name,  
c.category_name  
from production.products p inner join production.brands b on p.brand_id=b.brand_id  
inner join production.categories c on c.category_id=p.brand_id;
```

Zadaci

Mogu li se isti upiti napraviti s `right join`?

- Prikažite sve proizvode koje nemaju pripadajuću kategoriju

```
select p.product_name,  
b.brand_name,  
c.category_name  
from production.products p inner join production.brands b on p.brand_id=b.brand_id  
left join production.categories c on c.category_id=p.brand_id  
where c.category_id is null;
```

- Prikažite sve proizvode (products) kojih nema ni u jednoj narudžbi (order_items)

```
SELECT product_name,  
order_id  
FROM production.products p LEFT JOIN sales.order_items o ON o.product_id = p.product_id  
WHERE order_id IS NULL
```


Self-join (rekurzivna veza)

- Prikažite popis zaposlenika (ime i prezime u jednoj ćeliji) i njihovih nadređenih (managera)

```
select  
e.first_name+' '+e.last_name zaposlenik,  
m.first_name+' '+m.last_name sef  
from sales.staffs e inner join sales.staffs m  
on m.staff_id=e.manager_id;
```

Je li na popisu glavni manager?

Agregatne funkcije

- Napravite upit koji će prikazivati prodaju po zaposleniku u kojem će se nalaziti:
 - Ime zaposlenika
 - Prezime zaposlenika
 - Godina narudžbe
 - Ukupna suma narudžbi zaposlenika

COUNT()

SUM()

AVG()

MIN()

MAX()

	ime	prezime	godina	iznos
1	Mireya	Copeland	2016	273519.90
2	Layla	Terrell	2016	136280.91
3	Venita	Daniel	2016	955812.27
4	Kali	Vargas	2016	163126.81
5	Genna	Serrano	2018	277870.04
6	Genna	Serrano	2017	319427.05
7	Layla	Terrell	2018	62191.60
8	Venita	Daniel	2017	1236370.12
9	Marcelene	Boyer	2017	1528096.08
10	Layla	Terrell	2017	247433.08
11	Kali	Vargas	2017	203533.40
12	Marcelene	Boyer	2018	585473.24
13	Mireya	Copeland	2017	310655.29
14	Mireya	Copeland	2018	253248.46
15	Venita	Daniel	2018	695171.09
16	Marcelene	Boyer	2016	825319.41
17	Kali	Vargas	2018	150034.96
18	Genna	Serrano	2016	355425.17

Primjer

- Pogledajte zadaću *2. zadaća (osnovni SQL upiti)* i preuzmite datoteku `v_2.sql` sa zadacima
- Rješenja zalijepite u istu datoteku, promijenite naziv u *PrezimeIme_v2* i predajte do 19.03.